Noise and Outliers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Noise and Outliers before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Overfitting Intuition

Without opening the old lesson, explain one key idea from Over**fitting** Intuition. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Decision Trees

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Training and Testing

Without opening the old lesson, explain one key idea from **Training** and Testing. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

Complete the supported Noise and Outliers trace and record the first step you can explain without help.

Use W2: Compute two **median**s for **Mean median** and **unusual value**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Noise and Outliers explanation against the supplied facts before discussing it.

Use the worked case for **Mean median** and **unusual value**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Compare summaries independently

For 1, 2, 2, 3 and 12, calculate **mean** and **median**. Then replace 12 with 2 and recalculate. Explain the change.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Noise and Outliers target to a trace and a justified conclusion.

Page counts are 72,80,96,960. The last book may be an anthology. Should a cleaning script delete it automatically? Give a defensible investigation and document the outcome.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Noise and Outliers case and show the deciding evidence.

Recorded lengths are 90,95,100,105,900 pages.

> Is 900 automatically wrong?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Page counts are 40, 42, 43 and 900. Should 900 be deleted immediately? Give two explanations to **investigate**.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 17 Test and reflect

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Noise and Outliers adds to Transparent Book Recommender and keep the evidence.

Keep legitimate unusual books so the recommender can serve unusual readers.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Page counts are 72,80,96,960. The last book may be an anthology. Should a cleaning script delete it automatically? Give a defensible investigation and document the outcome.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Compute a mean

> Compute a mean. Apply the supplied Noise and Outliers facts and keep your reasoning.

Find the **mean** of 4, 4, 6, 6 and 10.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Compute two medians

> Compute two medians. Apply the supplied Noise and Outliers facts and keep your reasoning.

Find **median**s of [9, 1, 3] and [8, 2, 4, 6].

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Repair an explanation

> Repair an explanation. Apply the supplied Noise and Outliers facts and keep your reasoning.

A learner deletes 18 because it is above the **mean**. What **evidence** is needed before correcting or excluding it?

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Compare summaries independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Compare summaries independently. Apply the supplied Noise and Outliers facts and keep your reasoning.

For 1, 2, 2, 3 and 12, calculate **mean** and **median**. Then replace 12 with 2 and recalculate. Explain the change.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Report missing evidence

Report missing evidence. Apply the supplied Noise and Outliers facts and keep your reasoning.

A dataset is empty. Is reporting a **mean** of 0 justified?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Noise and Outliers

> A fictional library records five book-processing times in minutes. Explain the labelled step before checking the changed condition.

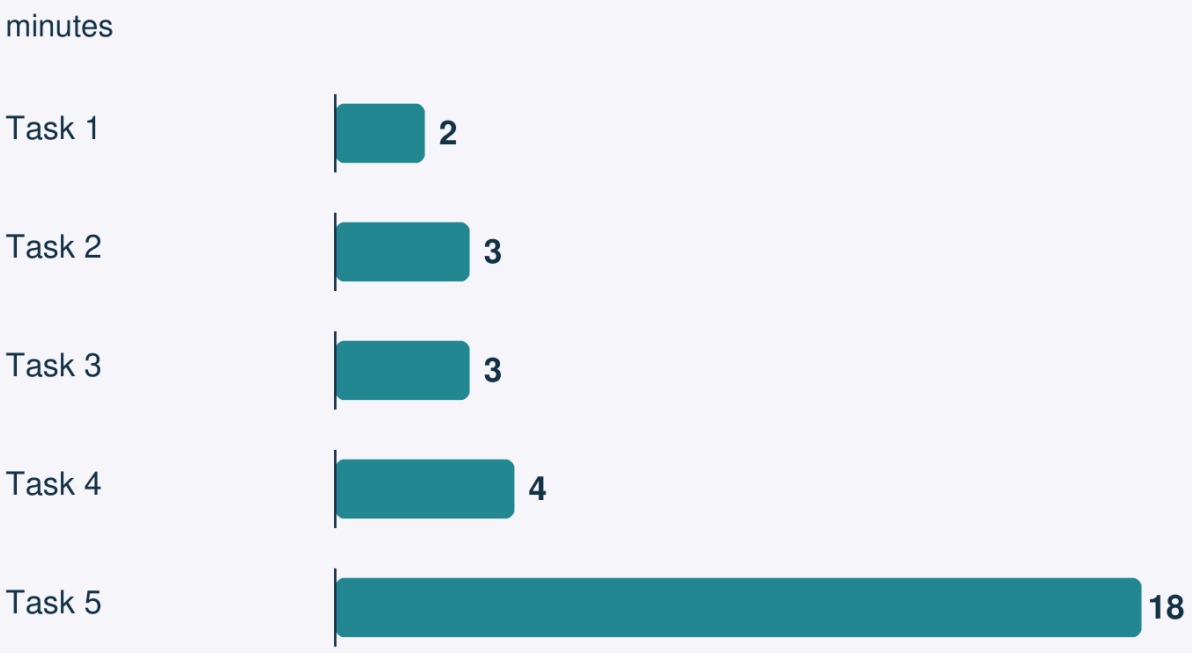
A fictional library records five book-**process**ing times in minutes.

Sorted times: 2,3,3,4,18.

All are recorded measurements; no **evidence** yet shows 18 is a mistake.

AI ACADEMY / AI-2/17

Retain unusual recorded measurements



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/17

Noise and Outliers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Mean

Sum 30; divide by 5: mean 6 minutes.



2. Median

The middle of five sorted values is 3 minutes.



3. Investigate 18

Check the source record and whether an unusually slow task occurred.



4. Document repair

Change a value only if evidence supports a correction; preserve the raw record.

CHANGE ONE THING

An authorised log confirms 18 was a transcription error for 3.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: **Investigate** unusual recommender features and show **evidence**-backed repairs instead of removing inconvenient records.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Investigate unusual recommender features and show **evidence**-backed repairs instead of removing inconvenient records.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional library records five book-**process**ing times in minutes.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Change a value only if **evidence** supports a correction; preserve the raw record.

> 1. **Median**

2. **Investigate** 18

3. Document repair

4. **Mean**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Document repair Change a value only if **evidence** supports a correction; preserve the raw record. The **mean** uses every value and is pulled upward by 18.

The **median** identifies the middle recorded time and stays at 3. Neither statistic declares the unusual **observation** invalid.

Deleting it simply because it changes the **mean** would hide potentially real slow work and change the question being answered.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: An authorised log confirms 18 was a transcription error for 3. Which conclusion is supported by the supplied facts?

1. Corrected **mean** is 3 minutes and **median** is 3.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Corrected **mean** is 3 minutes

and **median** is 3. Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

Repair a missing building block

Review Over**fitting** Intuition (ai-2/16). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: Does zero **training** error justify choosing Complex in the original record?.

Return to this week and explain what you changed.

Review Decision Trees (ai-2/14). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Why does changing colour alone not change this tree's **prediction**?.

Return to this week and explain what you changed.

Review **Training** and Testing (ai-2/11). Goal: Choose a split unit that matches the intended **prediction** claim. Short practice: How many distinct books are in each repaired role, and why report that as well as scan counts?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

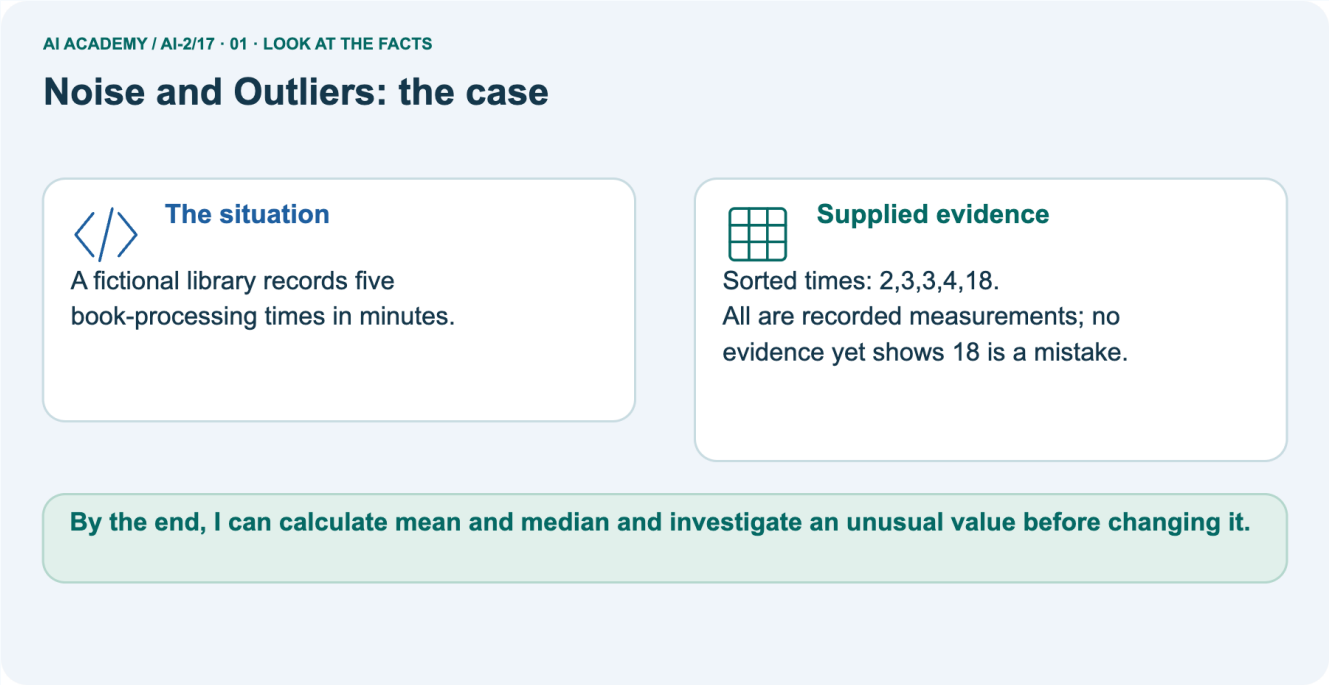
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Noise and Outliers picture from its labels, then record a relevant limit.

01. Noise and Outliers: the case



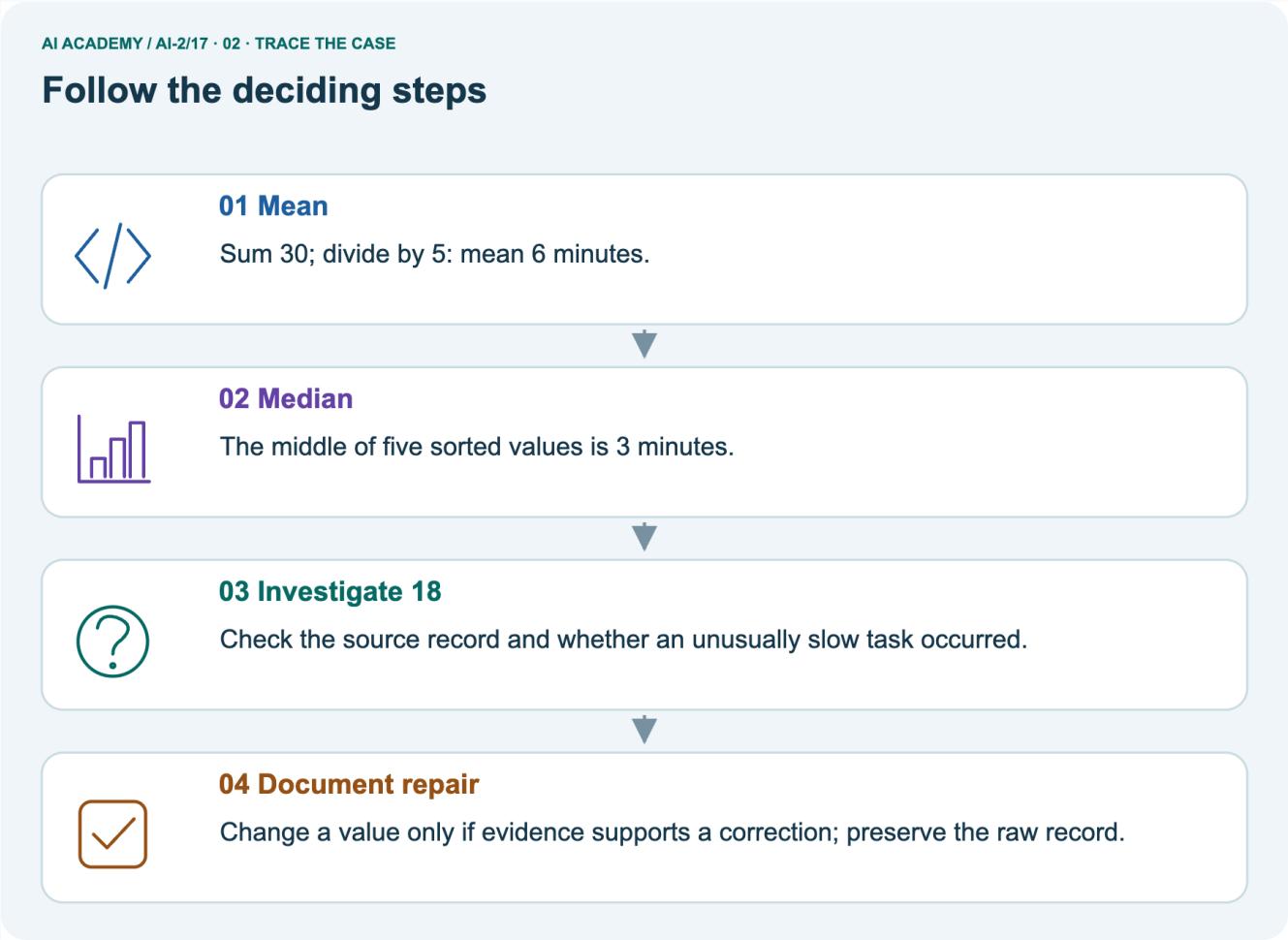
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional library records five book-**process**ing times in minutes. Supplied facts: Sorted times: 2,3,3,4,18. All are recorded measurements; no **evidence** yet shows 18 is a mistake.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



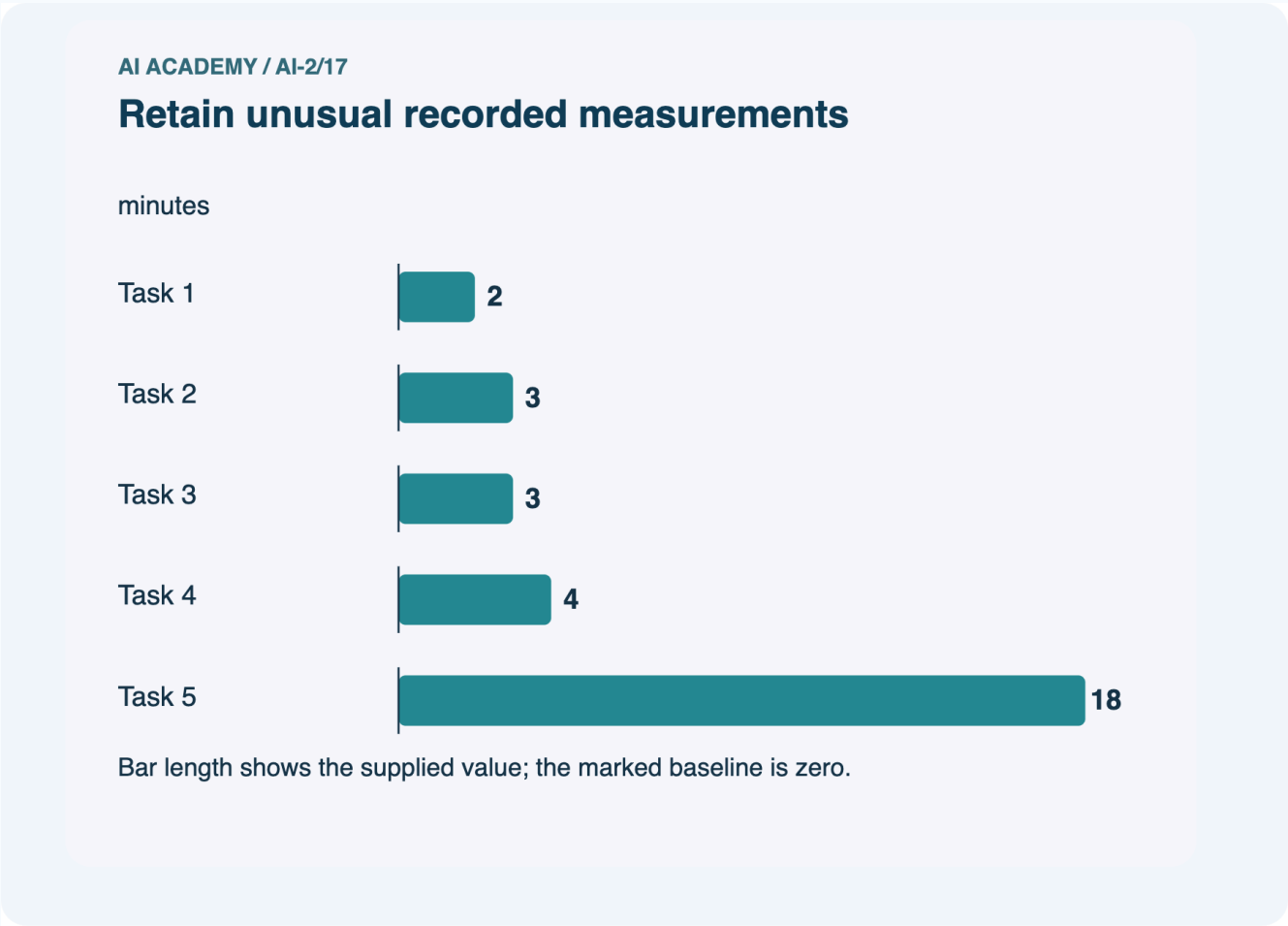
Follow the labelled picture with the matching explanation. Open the vector diagram

Mean: Sum 30; divide by 5: **mean** 6 minutes. → **Median**: The middle of five sorted values is 3 minutes. → **Investigate** 18: Check the source record and whether an unusually slow task occurred. → Document repair: Change a value only if **evidence** supports a correction; preserve the raw record.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Retain unusual recorded measurements



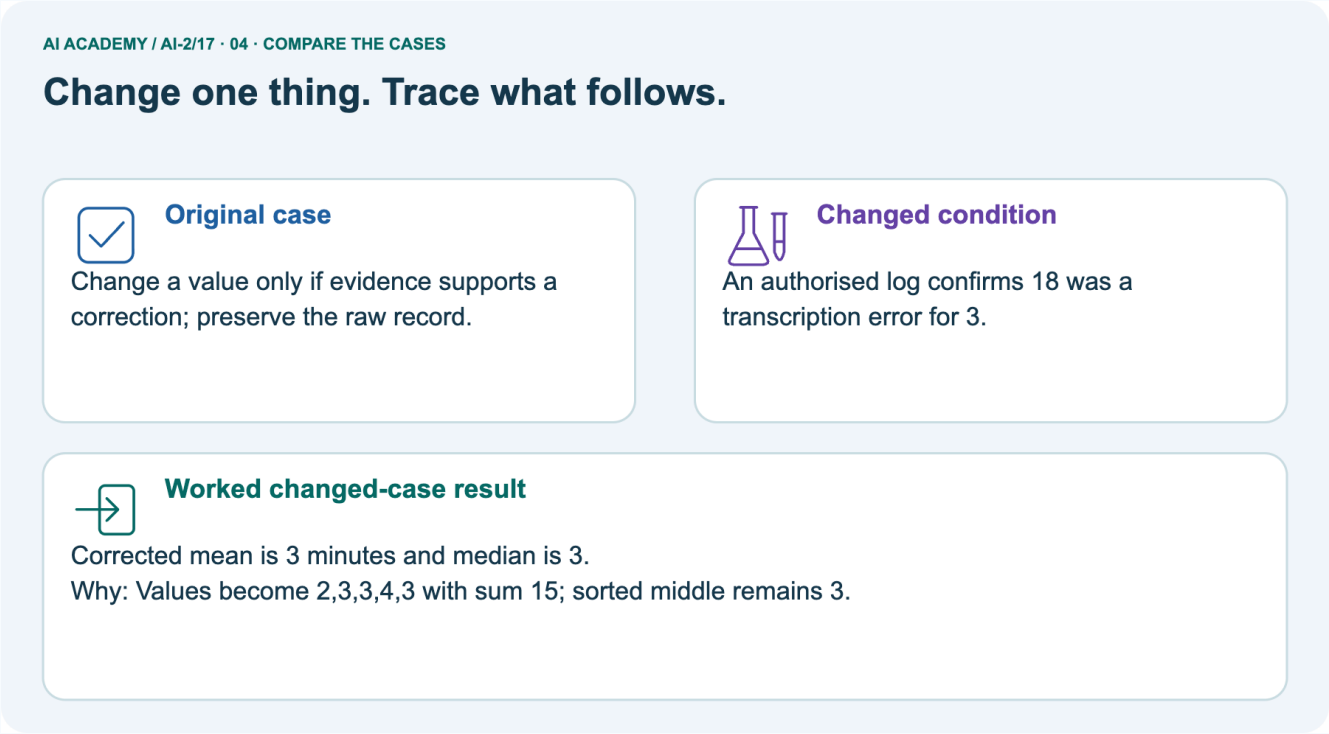
Follow the labelled picture with the matching explanation. Open the vector diagram

Retain unusual recorded measurements. Units: minutes. Task 1: 2. Task 2: 3. Task 3: 3. Task 4: 4. Task 5: 18.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

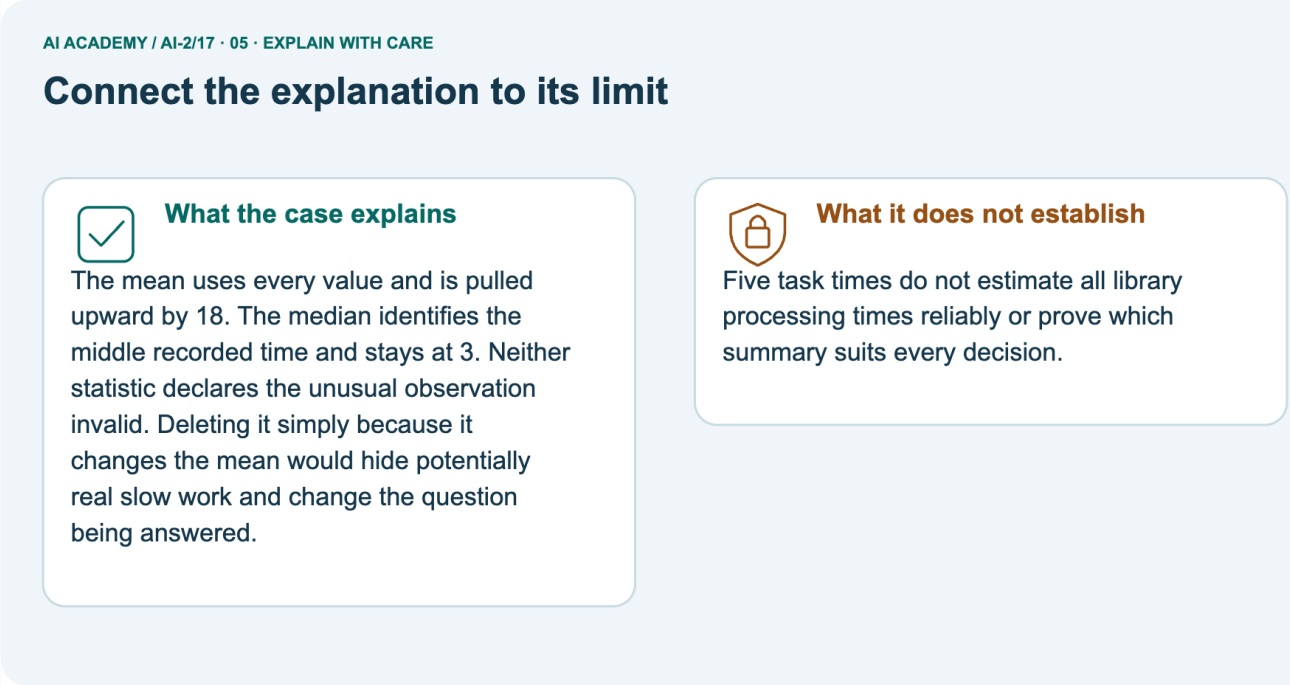
Original case: Change a value only if **evidence** supports a correction; preserve the raw record.
Changed condition: An authorised log confirms 18 was a transcription error for 3.

Expected result: Corrected **mean** is 3 minutes and **median** is 3. Reason: Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The **mean** uses every value and is pulled upward by 18. The **median** identifies the middle recorded time and stays at 3. Neither statistic declares the unusual **observation** invalid.

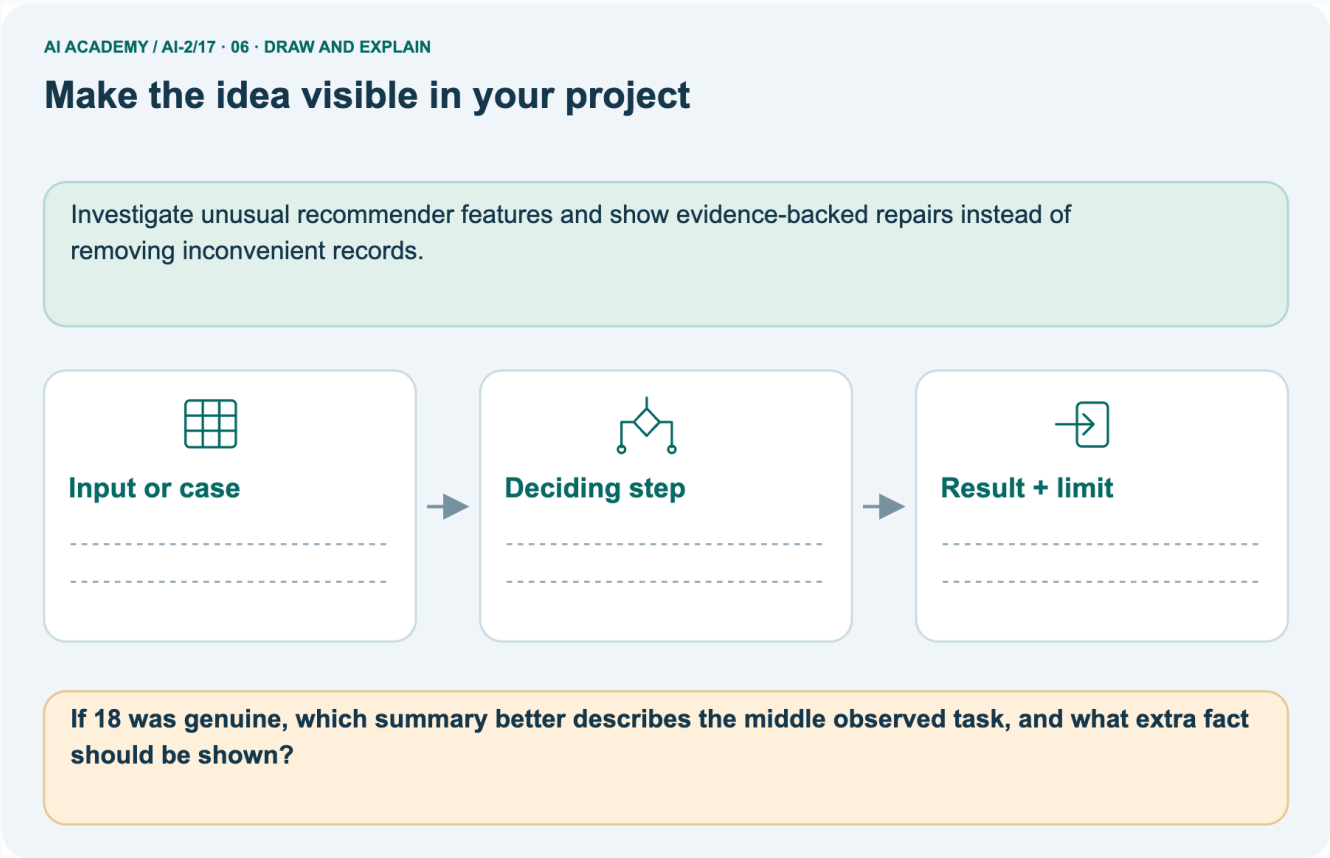
Deleting it simply because it changes the **mean** would hide potentially real slow work and change the question being answered.

Limit: Five task times do not estimate all library **process**ing times reliably or prove which summary suits every decision.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Investigate unusual recommender features and show **evidence**-backed repairs instead of removing inconvenient records. Draw your **input**, deciding step, result and limit.

Supported question: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.